

Dictionary L4 — masses from the engine v0.1 (framework): the lepton's Casimir $5/2 = \rho_1$ = the Bergman kernel singularity exponent $\rightarrow$ the lepton is the NATURAL mass-setting scale.

Mass mechanism is sector-dependent: fermion (Casimir-as-kernel-exponent) / gauge (EWSB) / scalar (VEV) / bulk composite (Bergman volume). The $6\pi^5 = C_2 \cdot \pi^{(n_C)}$ structural decomposition organizes the $e \rightarrow p$ bridge.

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Dictionary L4 — masses from the engine (framework v0.1)

0. Where this sits

The static 5-tuple is essentially derived (L1-L3 + engine v0.4); L4 starts the dynamic axis: mass scales. The mass-from-engine question is multi-faceted because mass mechanisms differ by sector (fermion $\neq$ gauge $\neq$ scalar $\neq$ composite). This v0.1 frames the structure and identifies the dictionary's key mass-setting alignment; explicit per-particle mass derivations connect to existing BST mass results in v0.2+.

1. The key alignment (rigorous): the lepton's Casimir IS the Bergman kernel singularity exponent

From the Casimir-anchoring (Grace #413 + the fundamental-sector selection):

Casimir(lepton K-type $(1/2, 1/2)$) = $5/2 = \rho_1$ = the Bergman kernel singularity exponent of D_{IV}^5 (= the Hua genus, the $n_C/\text{rank} = 5/2$ invariant).

This is not a coincidence — it's an alignment: the LEPTON (the fermion, the spinor fundamental, the matter sector) sits exactly at the substrate's natural mass-setting singularity. The Bergman kernel governs the Hilbert-space inner product on holomorphic functions; its singularity exponent sets the natural mass scale of the substrate's discrete-series spectrum.

The lepton mass is the substrate's natural mass unit. All other mass scales are ratios from it, set by the engine's structure on each sector.

This is the dictionary's organizing principle for mass: the lepton is the natural unit because its K-type Casimir equals the kernel exponent. Matter sets the scale.

2. Mass mechanism varies by sector (framework)

sector	K-type	Casimir	mass mechanism	dictionary contribution
Higgs (scalar)	(0,0)	0	VEV (Higgs mechanism)	trivial Casimir → no intrinsic mass; mass comes from EWSB
lepton (fermion)	(1/2,1/2)	$5/2 = \rho_1$	Casimir-as-kernel-exponent	natural mass-setting scale (the dictionary's anchor)
photon (gauge)	(1,0)	4	gauge symmetry → massless	Casimir is structural label (4 = rank ²), not mass
W/Z (gauge adjoint)	(1,1)	$6 = C_2$	EWSB (Higgs mechanism)	$m_W/m_Z = \sqrt{g/N_c}$ (Weinberg, derived) — RATIO from substrate primaries

So the dictionary handles each sector's mass mechanism appropriately: fermions get a NATURAL scale from the kernel-exponent alignment; gauge bosons get RATIOS from substrate primaries (Weinberg); scalars get mass via EWSB on the trivial Casimir.

3. The cross-sector bridge: $6\pi^5 m_e = m_p$ decomposes as $C_2 \cdot \pi^{(n_C)}$

Casey's existing result: $m_p = 6\pi^5 m_e$ (0.002% precision) — the proton mass from the electron mass via a $6\pi^5$ factor. The dictionary's structural decomposition:

$$6\pi^5 = C_2 \cdot \pi^{(n_C)}$$

where $C_2 = 6$ is the second Casimir invariant of B_2 (Grace's adjoint K-type Casimir) and $\pi^{(n_C)} = \pi^5$ comes from the $n_C=5$ dimensional Bergman volume factor for D_{IV}^5 . So the $e \rightarrow p$ bridge factor decomposes cleanly into two substrate primaries (C_2 and n_C), one Casimir and one dimension.

This is a STRUCTURAL OBSERVATION (the $6\pi^5$ result is Casey's; the decomposition $6\pi^5 = C_2 \cdot \pi^{(n_C)}$ is arithmetic identifying which substrate primaries each factor is). It says the dictionary organizes the bridge as "lepton (fermion, natural scale) → proton (bulk composite, scaled by $C_2 \cdot \pi^{(n_C)}$ for the Casimir-times-volume jump from boundary to bulk)." Explicit derivation = v0.2.

4. The Weinberg gauge-sector ratios (existing derived results)

For the gauge sector, the dictionary directly produces the EWSB ratios: $-\sin^2\theta_W = \text{rank} / N_c^2 = 2/9$ (derived, Weinberg). $-m_W / m_Z = \sqrt{g / N_c}$ (derived).

So the gauge sector's mass ratios are dictionary outputs (substrate primaries → derived ratios). The dictionary doesn't set the absolute gauge scale (that comes from EWSB / the Higgs VEV, which sets the electroweak scale ~ 246 GeV); it sets the structural RATIOS.

5. What v0.1 delivers vs what's staged for v0.2

Delivered (framework): - The lepton-Casimir-as-kernel-exponent alignment (the dictionary's organizing principle for mass: the lepton is the natural unit). - Mass-mechanism-by-sector classification (fermion / gauge / scalar / composite). - Structural decomposition $6\pi^5 = C_2 \cdot \pi^{(n_C)}$ organizing the $e \rightarrow p$ bridge.

Staged for v0.2+: - Explicit derivation of the lepton mass from the Bergman kernel singularity (connects to Track BC, the hydrogen 1s Bergman integral evaluation done earlier). - Explicit derivation of the $6\pi^5$ factor as $C_2 \cdot \pi^{(n_C)}$ from the boundary→bulk inclusion mechanism. - Lepton mass ratios ($e/\mu/\tau$) from the K-type radial tower structure. - Quark masses (which depend on the bulk-composite structure + the dictionary's per-particle map, #416). - Generation-related mass hierarchies (tied to the open generation gate, #414).

6. Honest scope + tier

RIGOROUS (alignment, observation): lepton Casimir = $5/2 = \rho_1$ (computed); ρ_1 = Bergman kernel singularity exponent = Hua genus of D_{IV}^5 (established result); arithmetic decomposition $6\pi^5 = 6 \cdot \pi^5 = C_2 \cdot \pi^{(n_C)}$.

FRAMEWORK (this doc): the dictionary's organizing principle is "lepton = natural mass unit (kernel alignment); other sectors = sector-specific mechanism." Not a full mass derivation; a structural setup for connecting the dictionary to existing BST mass results.

STAGED: explicit per-particle mass derivations (lepton ratios, quarks, the $C_2 \cdot \pi^{(n_C)}$ mechanism) — v0.2.

RIDES ON: the keystone bet (K-types = particles) like the rest of the dictionary; the existing BST mass results ($6\pi^5$ from kernel structure; Weinberg from substrate primaries) which I am NOT re-deriving but connecting to via the dictionary.

Cal #27 / honesty: I am NOT claiming masses are derived from the dictionary in v0.1 — I am framing the mechanism (sector-dependent), highlighting the lepton-Casimir/kernel-exponent alignment as the dictionary's organizing principle, and identifying the $C_2 \cdot \pi^{(n_C)}$ structural decomposition of the existing $6\pi^5$ result. The actual mass derivations build on the existing BST kernel/Weinberg machinery, with v0.2+ connecting them explicitly. Framework setup, not derivation.

Routed: → Grace: your Periodic Table can carry a "natural mass unit = lepton" annotation (the kernel-alignment); the sector-mechanism column tells you which cell's mass comes from what mechanism. → Elie: any kernel-singularity / Bergman-volume computation feeds the v0.2 explicit derivation. → me: #416 per-particle dictionary layer (flip per-cell 5-tuples) is the parallel next step; v0.2 explicit mass derivations later.

— Lyra, Dictionary L4 v0.1 (masses framework). KEY ALIGNMENT (rigorous): the lepton's K-type Casimir = $5/2 = \rho_1$ = the Bergman kernel singularity exponent of D_{IV}^5 — the lepton is the NATURAL mass-setting scale of the substrate (matter sets the unit). FRAMEWORK: mass mechanism varies by sector — fermion (natural Casimir/kernel scale) / gauge (EWSB, derived ratios $m_W/m_Z = \sqrt{g/N_c}$) / scalar (VEV) / bulk composite (Bergman volume). STRUCTURAL DECOMPOSITION: the existing $6\pi^5 m_e = m_p$ result organizes as $6\pi^5 = C_2 \cdot \pi^{(n_C)}$ — C_2 (adjoint Casimir) times the $\pi^{(n_C)}$ Bergman-volume factor, cleanly organizing the boundary→bulk bridge. v0.1 = framework + alignment; v0.2 = explicit per-particle derivations connecting to existing BST kernel/Weinberg machinery. Rides on keystone bet.